

PHYS 3200 - Fall 2026

End of Topic Quiz 2

September 17, 2026

Name and ID #:

Instructions:

1. You may use:
 - A calculator.
 - Notes.
 - Writing utensil: pencil and eraser are best.
2. If you have a question about a problem (confused about the situation, need some missing piece of information, etc.), please raise your hand and ask!
3.

 for each part.
4. Credit will not be given if your answers are too messy or obscure to read. Remember: a grader who isn't squinting and spending extra time trying to decipher mysterious scratches is a happy grader.
5. This quiz must be turned in by the end of the class period.

1. A particle of mass m is released from rest at $t = 0$. Take the positive y -direction to be vertically downward. The particle experiences gravity and a linear drag force

$$\mathbf{f} = -b\dot{\mathbf{r}},$$

where b is a positive constant.

- (a.) Draw and label a free-body diagram for the particle after it is released. Use the diagram to write Newton's second law for the vertical motion in terms of $\dot{y}$ and $\ddot{y}$.

- (b.) Find the terminal velocity $\dot{y}_{\text{ter}}$. Briefly state the condition that defines terminal velocity.

- (c.) Define the time constant

$$\tau = \frac{m}{b}.$$

Using $\dot{y}_{\text{ter}}$ and τ , separate and integrate the equation of motion to find $\dot{y}(t)$. Your answer must satisfy

$$\dot{y}(0) = 0.$$

2. A particle has an initial horizontal velocity $\dot{x}(0) = v_{0x}$ and experiences a linear drag force. Its horizontal velocity is

$$\dot{x}(t) = v_{0x}e^{-t/\tau},$$

where $\tau = m/b$.

- (a.) Integrate $\dot{x}(t)$ to find the horizontal position $x(t)$. Use the initial condition $x(0) = 0$.

- (b.) Find the limiting horizontal position $x(\infty)$. Briefly explain the physical meaning of this result.